

ROCKY ARCHES

Waves often attack a headland on both sides. They can carve away the rock near sea level to form twin caves that eventually break right through the headland, creating an arch. Arches can also form when caves in hard rock layers break through to softer layers beyond.

▼ CURVED WONDER

Natural rock arches are rare because the rock usually disintegrates, but some arches have lasted for many centuries.



TOUGH SURVIVORS

Wherever the sea attacks a rocky coastline, the hardest rocks tend to survive the longest. These usually take the form of long ridges that turn into headlands, but sometimes lumps of extra-hard rock survive as islands. These islands off Brazil, called the Two Brothers, are made of volcanic basalt rock that welled up from deep within the Earth long ago and hardened. It has resisted the waves ever since.



SEA STACKS

Usually, headlands under attack from the waves crumble into heaps of rubble. But in some places, columns of extra-hard rock survive as sea stacks. Cut off from the shore, and with sheer cliff faces on all sides, these make ideal nesting sites for seabirds. Over time, most fall into the sea, leaving behind rocky stumps that may be submerged at high tide.

◀ CRUMBLING SEA STACK

Gaps in the rock near sea level may mean that this sea stack is close to falling.